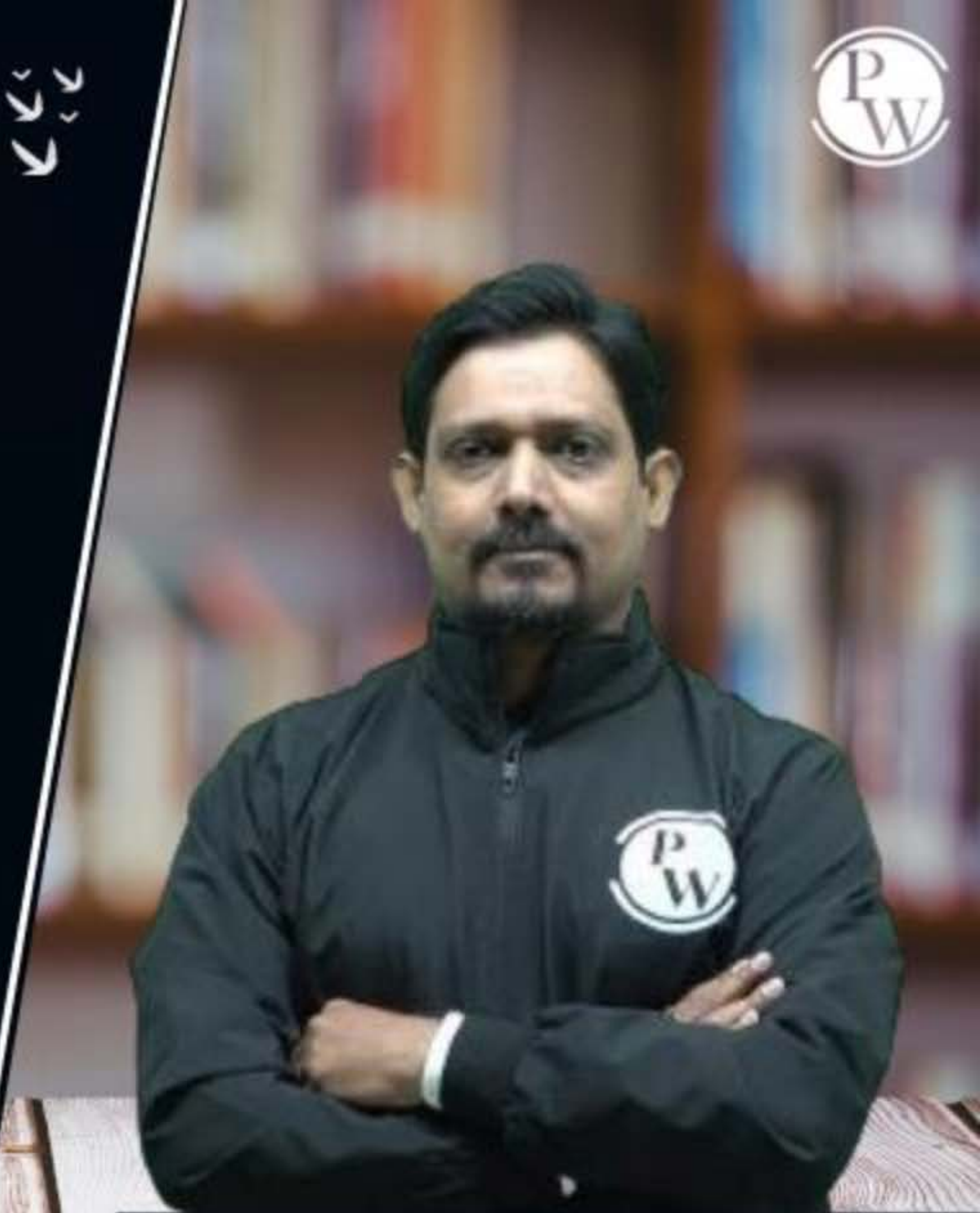
# Computer Science & IT

## Fundamental of C Language

### Array & Pointers

**Lecture No. 02**



By- Abhishek Sir

# Recap of Previous Lecture

**Topic** pointer variable

**Topic** * operator dereference operator

**Topic** & operator (Address of operator)

**Topic**

**Topic**

# Topics to be Covered

**Topic** Array data type

**Topic** Declaration of array

**Topic** Initialization of array

**Topic** How to access element of array

**Topic**

# Array

We are required to store 50 students Roll No in class

one way is declare 50 integer variable

int RN01, RN02, RN03 . . .

disadvantage :- We created Several variable does not represent the single logical.

(2) Managing all the variable may be difficut.

(3) The code size will increase

# Array

- * Another way is to declare a Single variable holding 50 values

- * Array is a variable that hold multiple values

- * Array is derived data type. (derived from primitive data type)
  int, char, float

- * Array is collection of Similar Data type

Slide

# Array

* Similary Means

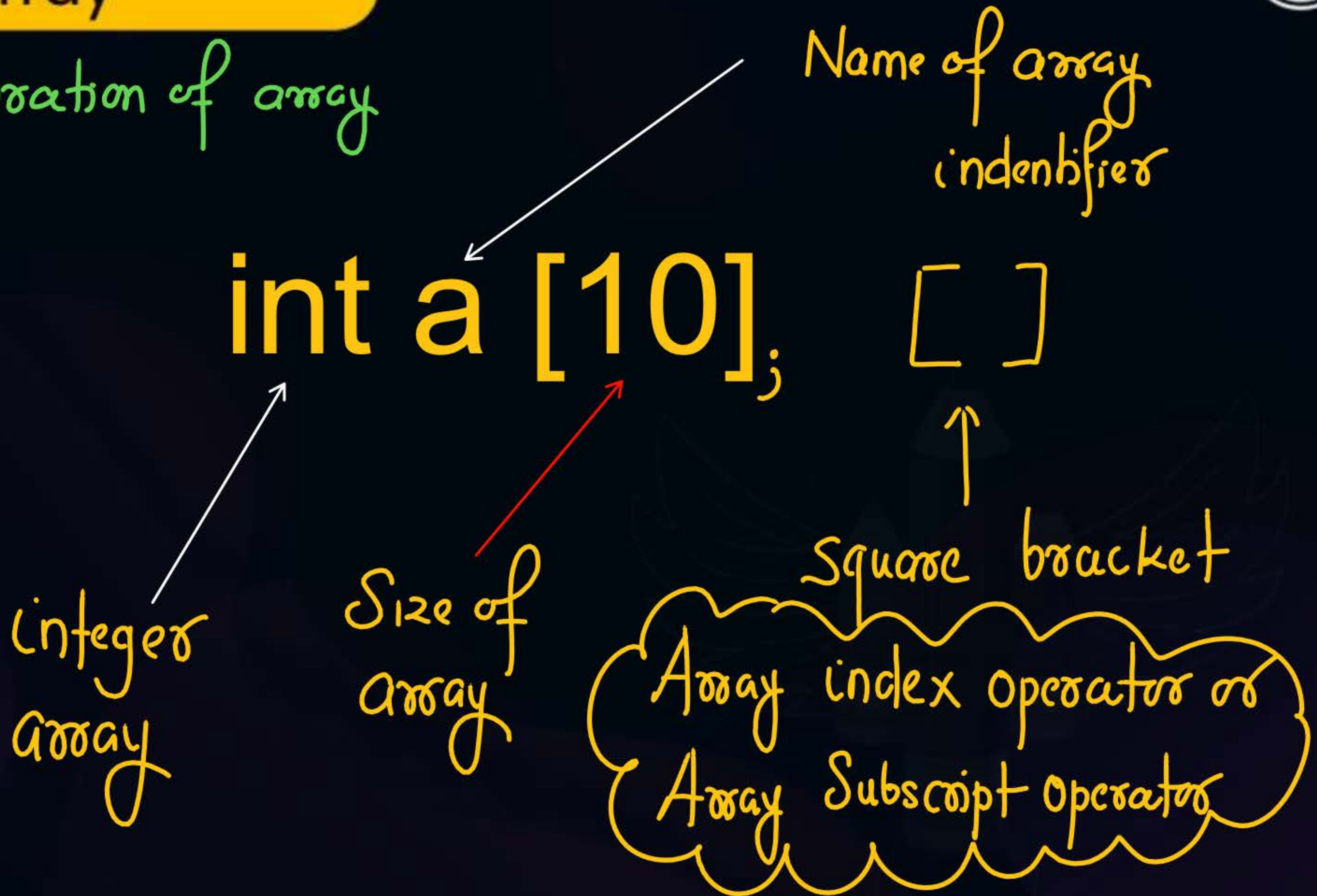
| | |
|---|---|
| all integer | integer array |
| all character | character array |
| all float | float array |

* Array store element in consecutive memoy Location

Consecutive means one after another.

# Array

Declaration of array



int a [10];

Name of array

indentifier

[ ]

integer array

Size of array

square bracket

Array index operator or
Array Subscript operator

# Array

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |

Initialization `int a[10] : {1,2,3,4,5,6,7,8,9,10};`

# Array

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| a | b | c | d | e | f | g | h | i | j |

Initialization char a[10] = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j'};

## Actually char ASCII

# Array

BA will be

$\boxed{1000}$

↑

assumed

value

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| a[0] | a[1] | a[2] | a[3] | a[4] | a[5] | a[6] | a[7] | a[8] | a[9] |

1000

Access array element

To access first element of array a[0]

`int a[10]; // first element a[0] .... a[9]`

# Array

The first element index is zero

```
printf("%d", a[0]);
printf("%d", a[1]);
   :
   :
printf("%d", a[9]);
```

The address of first element of array is called Base Address

# Array

Dimension means Size

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | ~~5~~<br>20 | 6 | 7 | 8 | 9 | 10 |

A[0] A[1] A[2] A[3] A[4]

```
# include<stdio.h>
int main() {
    int a[10] = {1,2,3,4,5,6,7,8,9,10};
    int i;
    a[4]=20; // update the array variable
    for ( i=0; i<=9; i++)
            printf("%d", a[i]);
    return 0.
}
```

Slide

# 2 mins Summary

**Topic** Array data type

**Topic** Array declaration

**Topic** Array Initialization

**Topic** Access/update

**Topic** Array Index

Array Dimension

Size

`int a[10];`

↑ 1-Dimension Array

Slide

# THANK - YOU